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Lecture #9: □□

Offered: 2026-04-09

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Outline

- Introduction to AI
- Adversarial attack & defense
- Model extraction
- Backdoor attack & defense
- Privacy
- DeepFake

Adversarial attack & defense

- Adversarial attack
  - White-box attack
  - FGSM,PGD,CW, JSMA ,Deepfool
  - Black-box attack
  - Transfer-based
- Adversarial defense
  - Adversarial training
- Robustness Evaluation

Adversarial attacks

- Formal definition

Caveat: formally defining an adversarial attack is helpful for mathematicians, but can hide some important real-world considerations

Real attackers don't care about your definitions

relation: $R(x, x')$

original image

altered image

speed limit: 45

what is ϵ?

example: $R_\infty(x, x') = ||x - x'||_\infty$

each pixel changed by at most ϵ

attack: $x^\star \leftarrow \arg\max_{x':R(x,x')\leq\epsilon} \mathcal{L}_\theta(x', y)$

e.g., $\mathcal{L}_\theta(x, y) = -\log p_\theta(y|x)$

pick image x' close to x

maximize the loss of your choice

defense: $\theta^\star \leftarrow \arg\min_\theta \underbrace{\sum_{(x,y)\in D} \max_{x':R(x,x')\leq\epsilon} \mathcal{L}_\theta(x', y)}_{\text{robust loss}}$

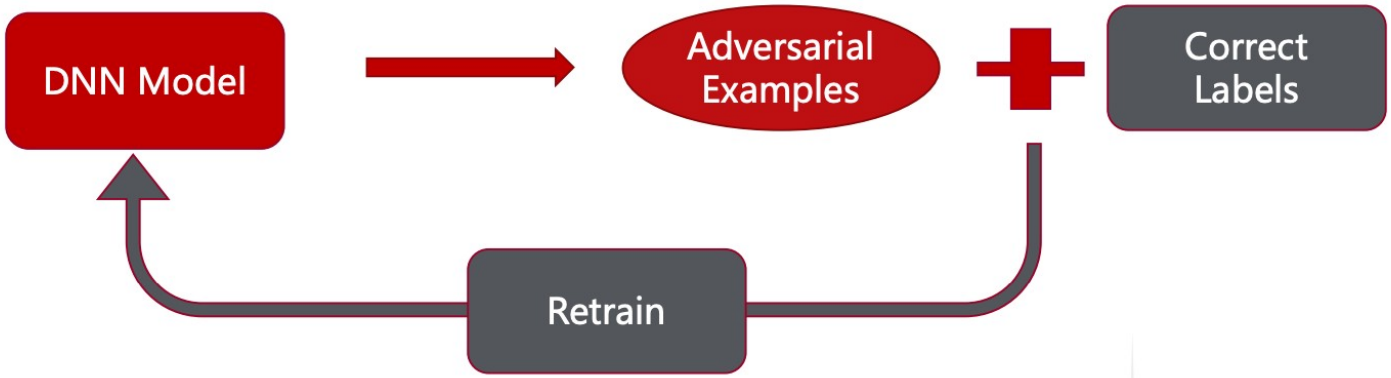
Adversarial attacks

- Attacks

Type	Algorithm
Score based attack	One pixel attack
Patch attack	Lavan, Dpatch
Gradient attack	FGSM, BIM, JSMA
Decision attack	Hopskipjump attack, Deepfool attack
Adaptive Attack	BPDA

Adversarial Defense

- Adversarial training



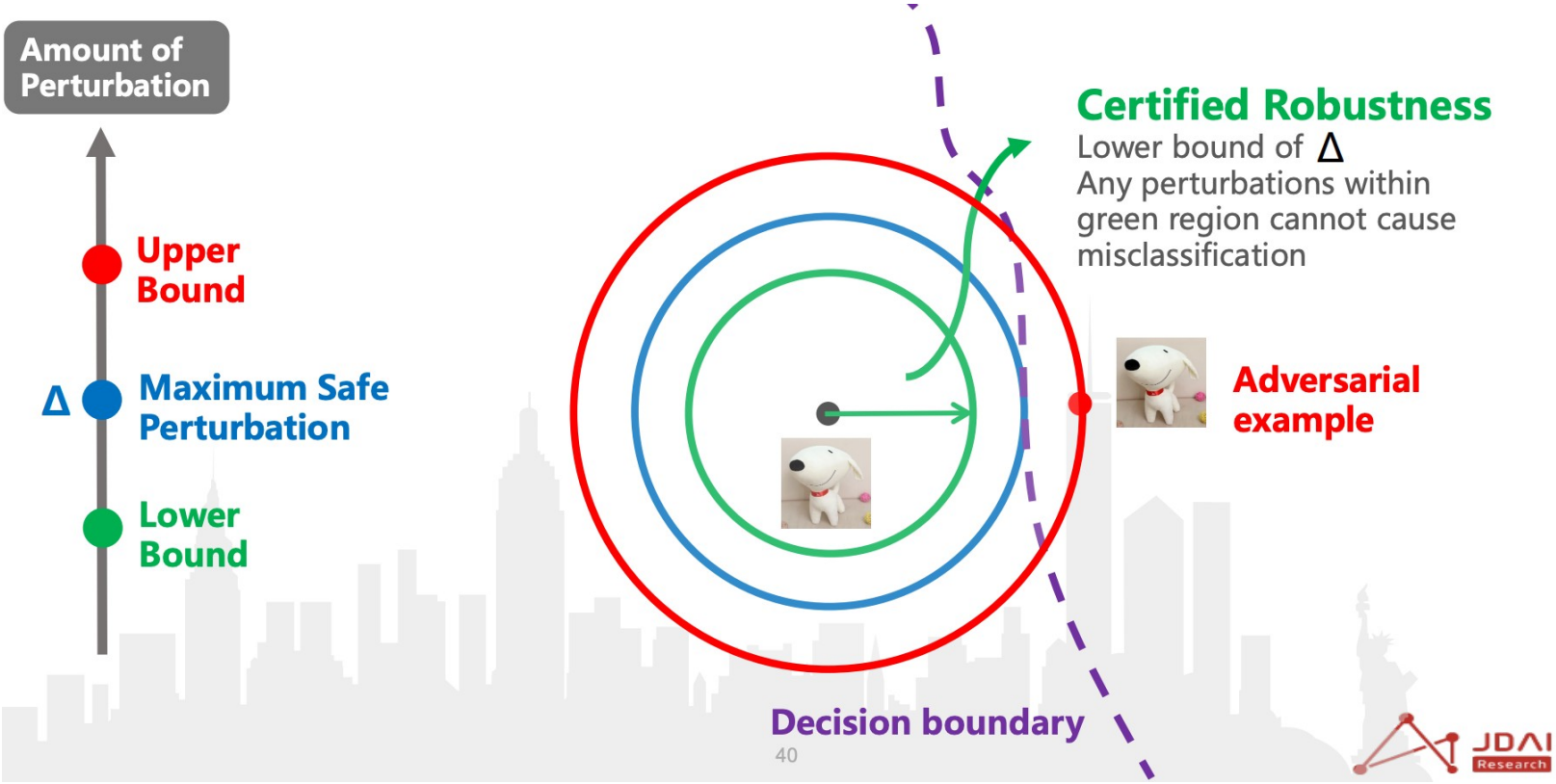
- Add adversarial examples to the training set, with their correct labels
- Robust optimization problem:

$$\theta^* = \arg\min_{\theta} \mathbb{E}_{x \in \mathcal{X}} \left[\max_{\delta \in [-\epsilon, \epsilon]^N} \ell(x + \delta; F_\theta) \right]$$

- (One of the) strongest defense mechanism so far

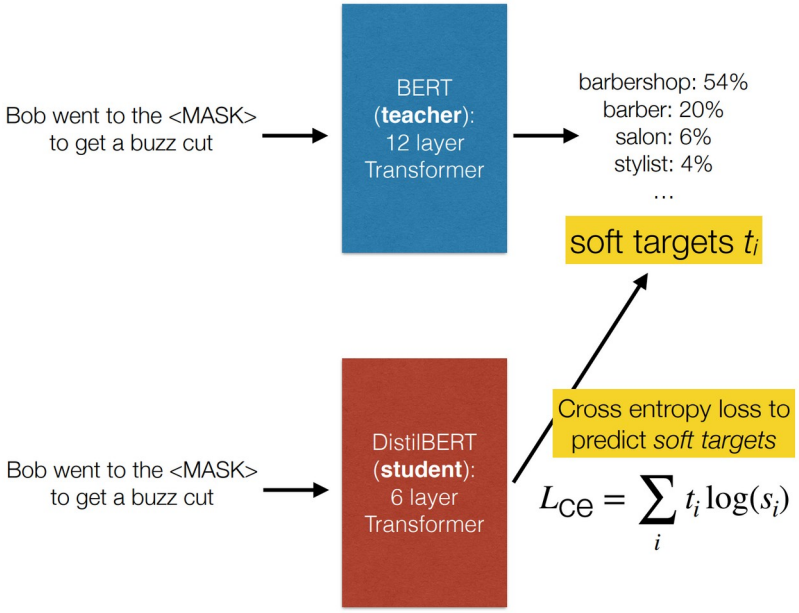
Adversarial Robustness Leaderboard

- Robustness Evaluation



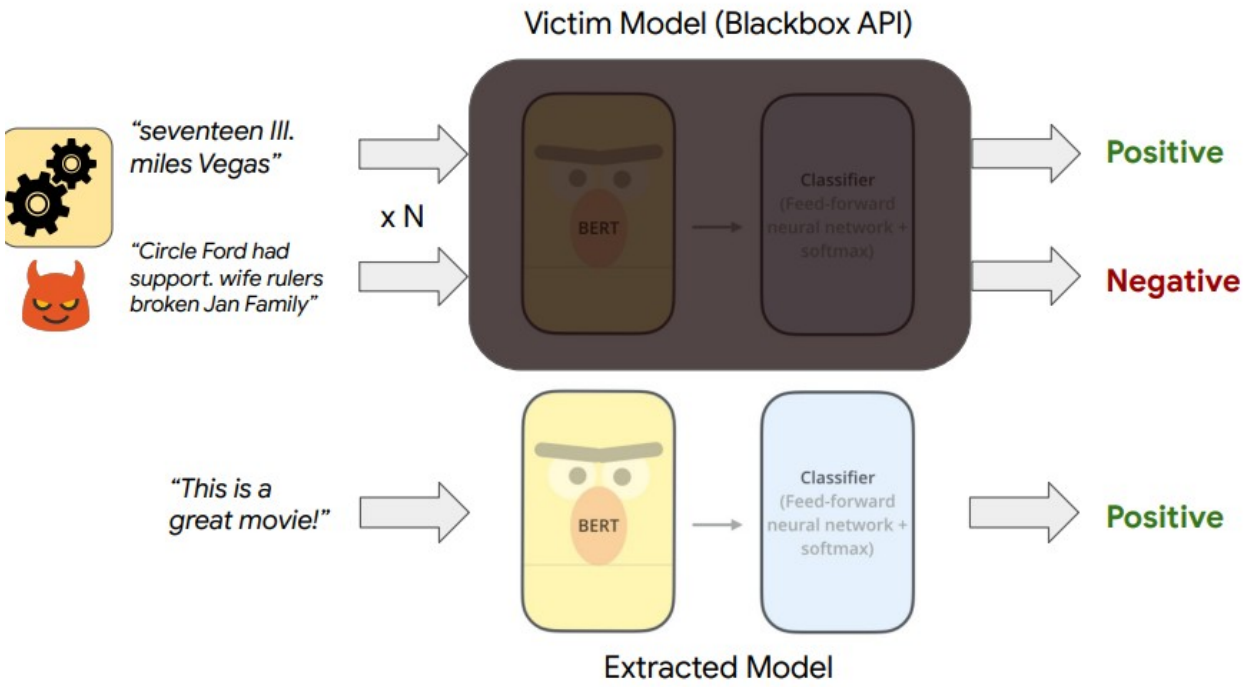
Adversarial attack & defense

- Model extraction
- Model Distillation
- Defense against extraction



Model Extraction

- The stolen copy works well on real data

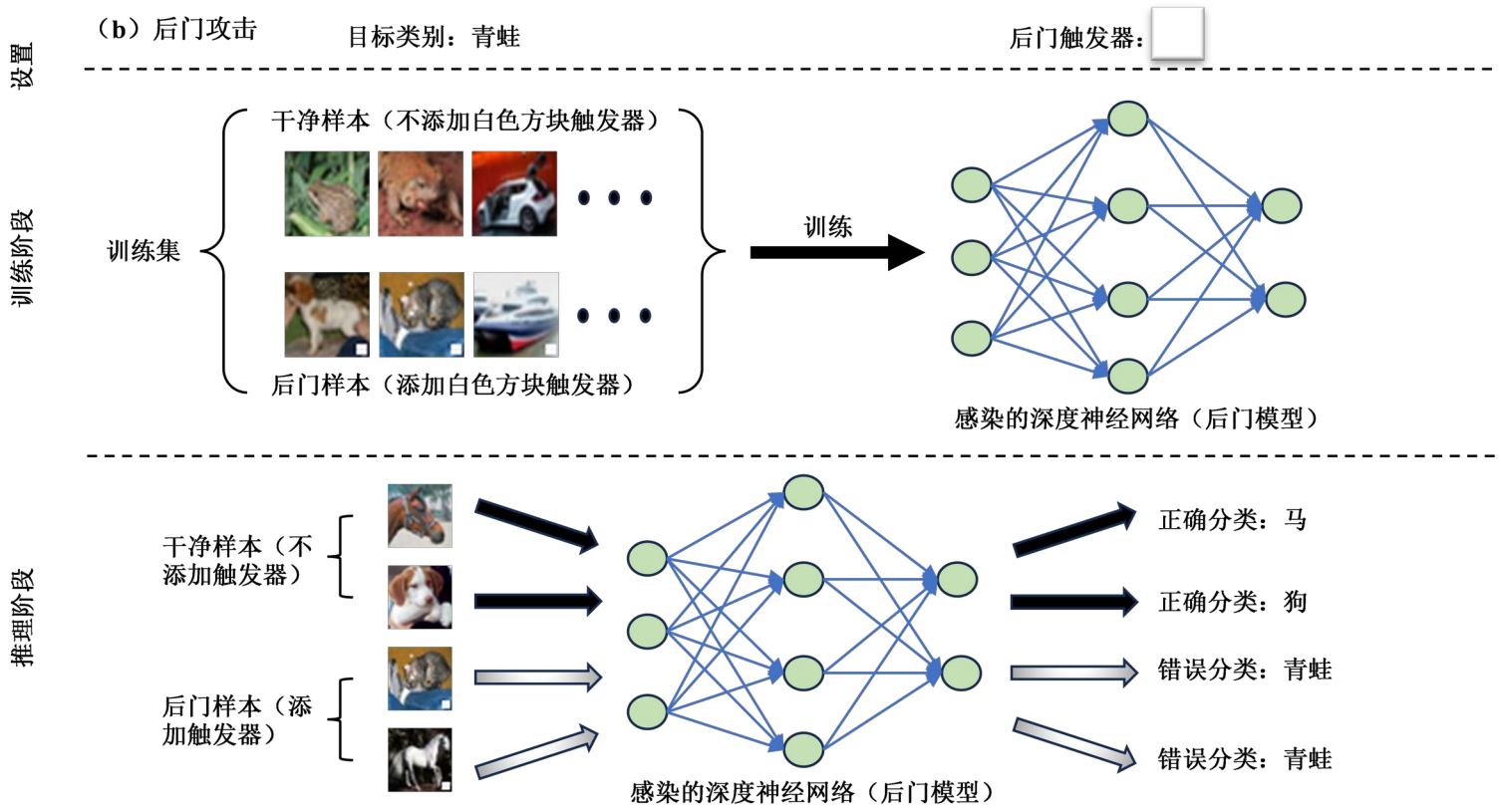


Backdoor attack & defense

- Backdoor attacks in AI
- Backdoor attacks taxonomy
  - Outsourcing, pretrained, data collection, collaborative learning, post-deployment,
  - code poisoning attack
- Model poisoning attacks in federated learning
- BadNets
- WaNet
- Clean-label attack
- Trojan attack

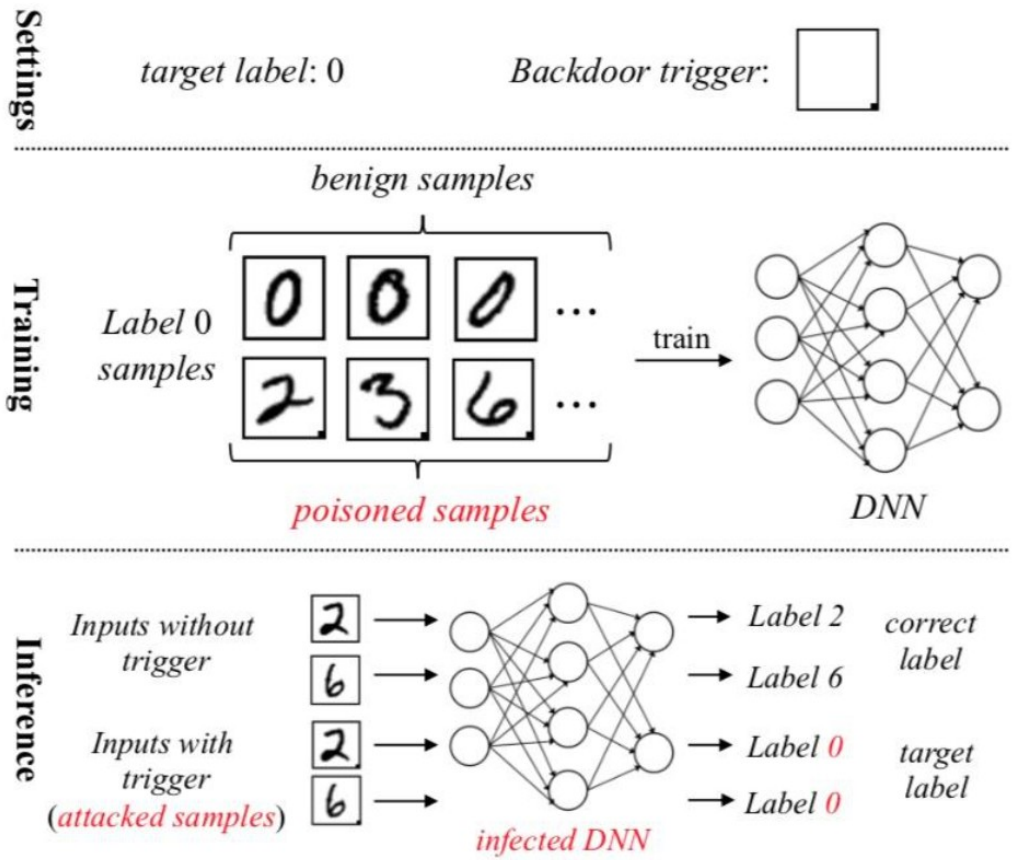
Overview

- Backdoor Attack:



Backdoor attacks

- BadNets:



Backdoor attacks

- WaNet:
  - Imperceptible warping-based backdoor attack

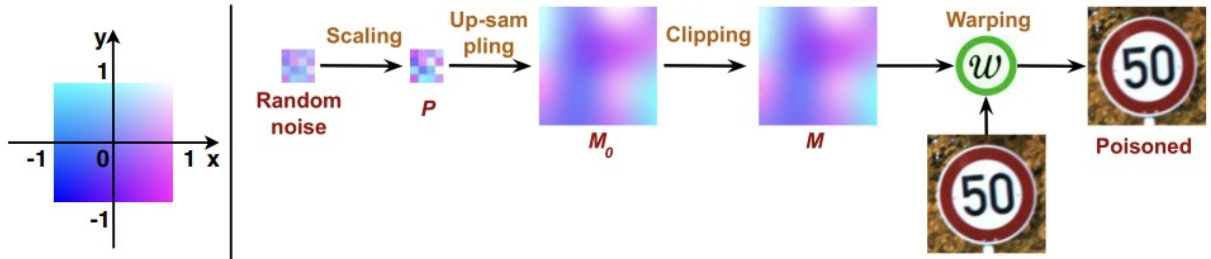
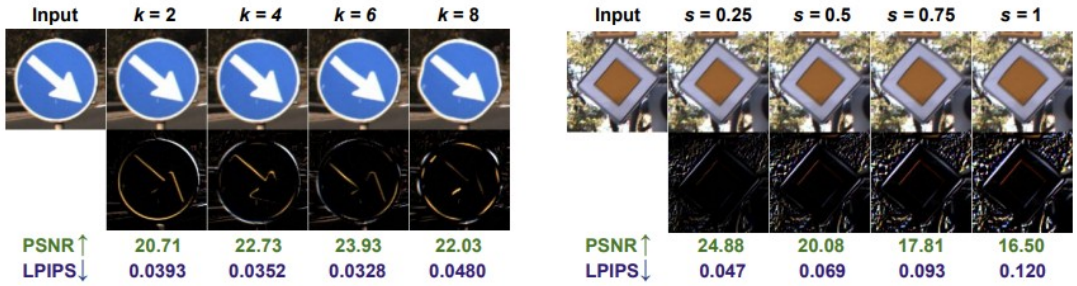


Figure 2: Process of creating the warping field M and using it to generate poisoned images.

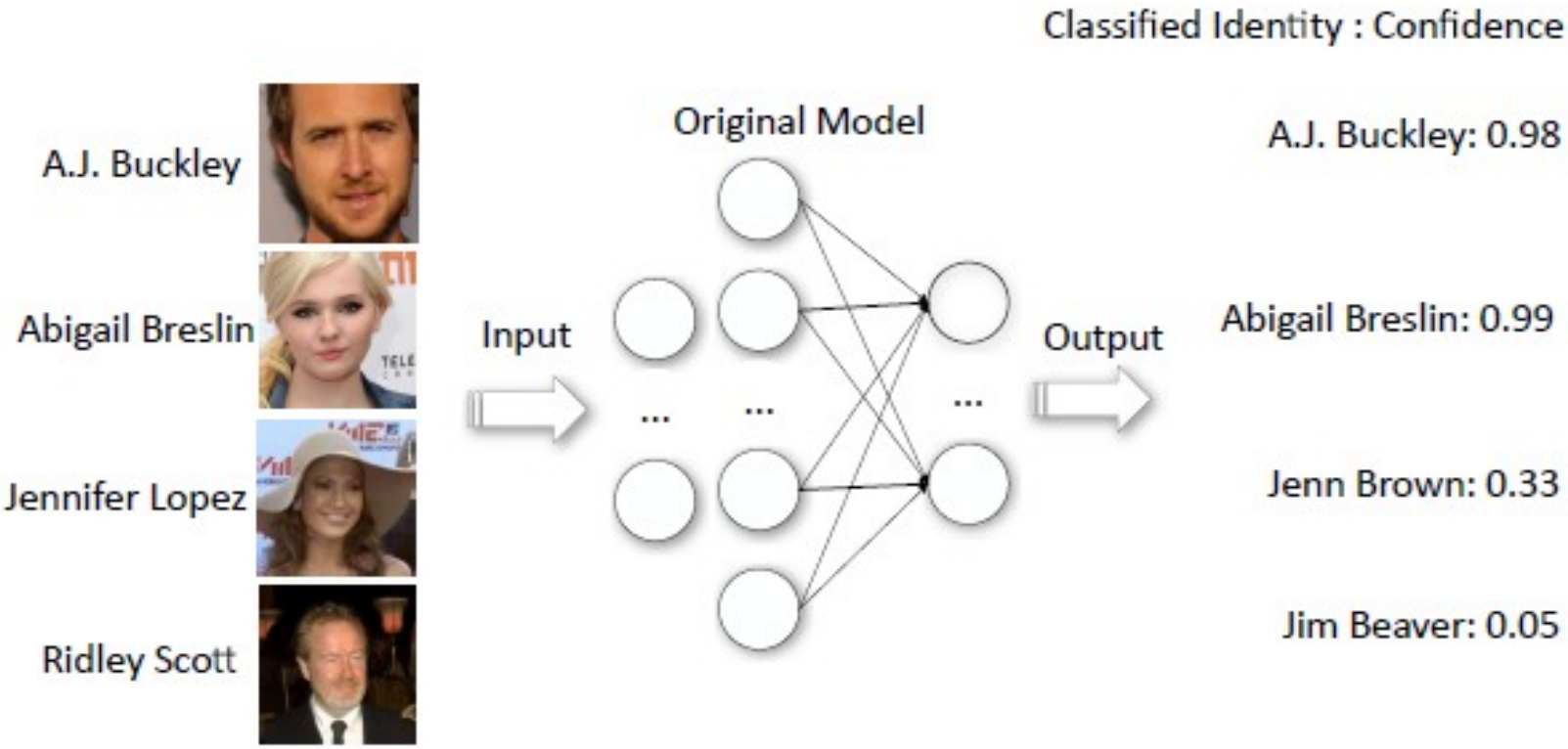


(a) Changing control-grid size k ($s = 0.5$) (b) Changing warping strength s ($k = 4$)

Figure 3: **Effect of different hyper-parameters on the warping result.** For each warped image, we show the image (top), the magnified ($\times 2$) residual map (bottom). The PSNR and LPIPS (Zhang et al., 2018) scores are computed at resolution 224×224.

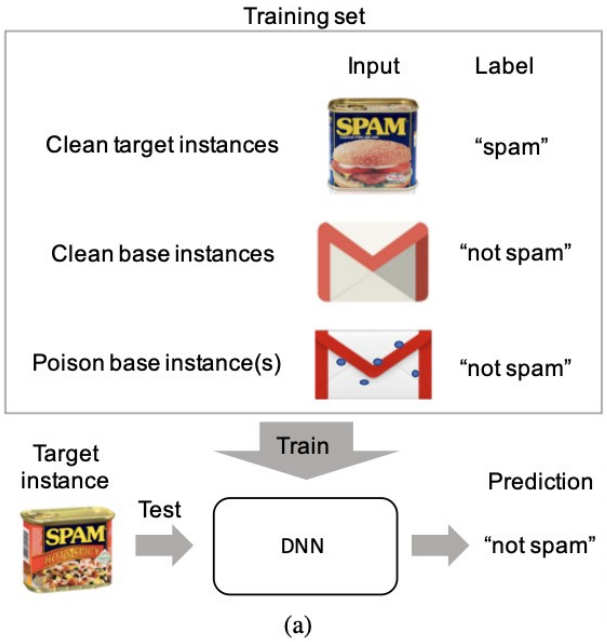
Backdoor attacks

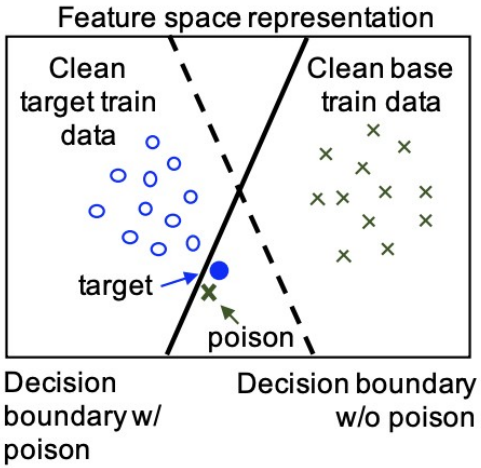
- Trojan attack:
  - Deep feature space



Backdoor attacks

- Clean-label attack:
  - Poison frogs





(b) Illustration of the feature space (activations of the penultimate layer before the softmax layer of the network) representation of clean training data, poison instance, and target instance. Note that the target instance is *not* in the training set. Thus it will not affect the loss when the nearby poison instance causes the decision boundary to shift to encompass both of them into the base region.

Backdoor defense

- ABL:

Anti-Backdoor Learning: Training Clean Models on Poisoned Data

- The defender’s goal:
- Training a model on the given dataset (potentially poisoned) that is as good as models trained on purely clean data.
- if an isolation method is used, the defender may identify only a subset of the backdoor examples

Backdoor defense

- AIBD

Adversarial-Inspired Backdoor Defense via Bridging Backdoor and Adversarial Attacks

Jia-Li Yin[1, 2], Weijian Wang[1, 2], Lyhwa[2*], Wei Lin[3, 4*], Ximeng Liu[2, 5]
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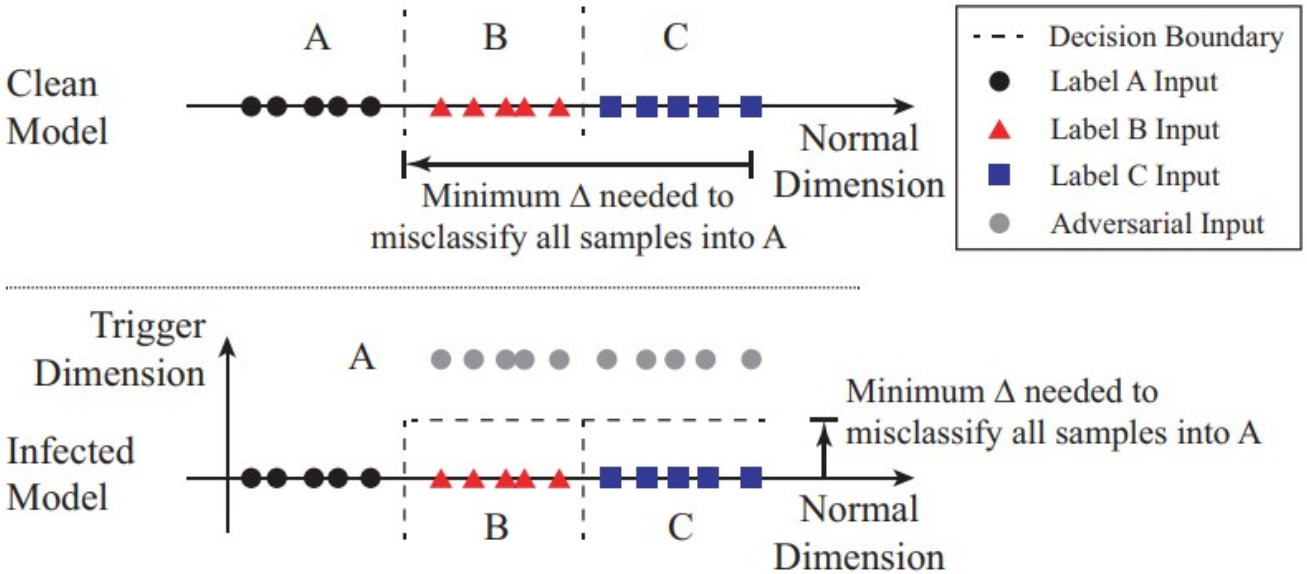
- While previous works generally treat these two attacks separately, the inherent connection between these two attacks is rarely explored.

Backdoor defense

- Neural Cleanse
- Detection

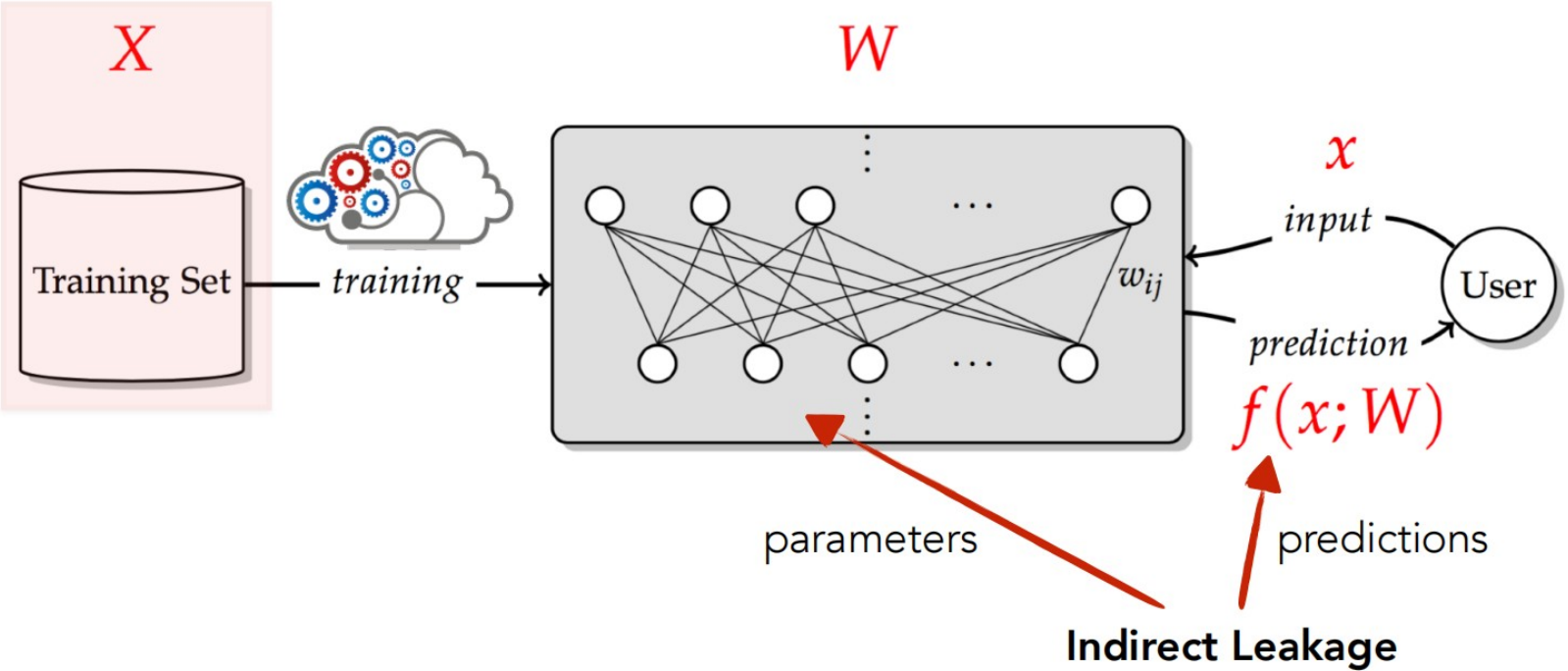


Backdoor attack & defense

- Backdoor defense in AI
- Backdoor defense taxonomy
- Anti-Backdoor Learning
- Adversarial-inspired backdoor defense
- Neural Cleanse

Privacy

- Privacy Risks in Machine Learning
- Membership inference attack
- Decentralized (Federated) Learning
- Neural Cleanse



Deepfake

- Deepfake
- Deepfake generation
  - Entire face synthesis
  - Attribute manipulation
  - Facial reenactment
  - Speaking manipulation
  - Face swap
- Deepfake Detection
  - Datasets
  - Image-based deepfake detection
  - Image-based deepfake segmentation
  - Video-based deepfake detection
  - Generalizability